The State Law for Lazy-Point Train Tracks: $f(N) = \min(2^N,\ N+4)$

Carl Heise*

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Abstract

A single train drives forever over a layout of N three-way switches with *lazy points*: entering at the stem, the train follows the current tongue and moves nothing; entering at a branch, it forces the tongue to that branch and leaves by the stem. How many distinct settings of the N tongues can one train ever exhibit? Exactly

$$f(N) \;=\; \min\bigl(2^N,\ N+4\bigr) \qquad \text{for every } N \geq 0.$$

The bound is uniform over all layouts, all starting positions, and all initial tongue settings; it is attained, for every N, by an explicit layout. The whole law is machine-checked as a single Lean 4 theorem, `GeneralN.state_law`, built without Mathlib, whose axiom audit is exactly the three standard Lean axioms. This document is a human-readable rendering of that proof: complete for the model, the 2^N ceiling, and every attainment construction, and a structured account of the sharp $N+4$ upper bound.

Contents

*Human-readable rendering extracted from the machine-checked Lean 4 development (`github.com/senegrom/duplotrain`, directory `theory/lean`); prepared with the assistance of Claude (Anthropic). The formal proof is the authority: every numbered statement below is a theorem of that development, and the sharp upper bound (Section 6) is presented as a faithful account of the formal argument rather than an independent proof.

1 Introduction

The question comes from a toy: LEGO® DUPLO® train track. Its Y-switches have unsprung, *lazy* points. A train entering the switch at its single stem is routed by whatever position the tongue happens to be in, and the tongue does not move. A train entering through one of the two branches pushes the points aside if necessary — the tongue ends up aimed at the branch the train came in by — and exits through the stem. A layout wires the free ends of switches to one another with plain track; plain track has no state, so the machine state of a layout is the vector of its N tongue positions, one bit per switch.

A single train, placed anywhere, either eventually falls off an unwired end or drives forever. Along the way the tongue vector changes. The *state count* of a run is the number of distinct tongue vectors it exhibits; $f(N)$ is the maximum state count over all N-switch layouts, all starts, and all initial tongue vectors. Trivially $f(N) \le 2^N$. The surprise is how far below 2^N the truth lies:

Theorem 1.1 (The state law)**.** *For every $N \ge 0$,*

$$f(N) \;=\; \min\bigl(2^N,\; N+4\bigr).$$

That is: no run on any N-switch layout ever exhibits more than $\min(2^N, N+4)$ distinct tongue vectors, and for every N some layout, start, and initial setting exhibits exactly that many.

One train can barely move the machine it drives over: of the 2^N conceivable switch settings, all but $N + 4$ are unreachable within any single run, no matter how the track is laid. The 2^N leg of the minimum binds only for $N \le 2$ ($f(0) = 1$, $f(1) = 2$, $f(2) = 4$); from three switches on, $f(N) = N + 4$.

The result was found and proved formally. A sequence of machine-checked bounds

$$26N{+}3 \to 24N{+}5 \to 18N{+}3 \to \cdots \to 2N{+}9 \to N{+}7 \to N{+}6 \to N{+}5 \to N{+}4$$

was tightened until the constructive lower bound was met. The final development (about 28,500 lines of Lean 4 in 70 files, using no external mathematics library) proves the single statement

$$\texttt{GeneralN.state\_law} \;:\; \forall N,\, \texttt{IsExactStateCount}\; N\; (\min(2^N, N+4))$$

and audits it to exactly the axioms $\{\texttt{propext}, \texttt{Classical.choice}, \texttt{Quot.sound}\}$ — in particular with no appeal to native code evaluation: the few finite computations are checked by the proof kernel itself.

What this document is. Sections 2–5 are complete human mathematics: the model, the ceiling $f(N) \le 2^N$, the resolution of the minimum, and all attainment constructions, each of which a reader can verify by hand. Section 6 renders the sharp upper bound $f(N) \le N + 4$ — by far the deep half, roughly 25,000 formal lines — as a structured account: the objects, the main lemmas, and the charging argument, with enough proof for each step that the shape of the mathematics is visible, and with pointers into the formal text, which remains the complete reference. Section 8 records what was formalised and how to check it.

The physical-trace core of the argument (Section 6.2) descends from the first-repeated-track idea in the train-set literature; the formal sources name Chalcraft–Greene and Aaronson.

2 The model

Everything is stated over a discrete dynamics in which plain-track geometry provably never enters; only the wiring pattern matters.

Definition 2.1 (Ports and wirings)**.** Switch k ($k = 0, 1, 2, \dots$) owns three *ports*, encoded as natural numbers: its *stem* $3k$, its *left branch* $3k + 1$, and its *right branch* $3k + 2$. A *wiring* is a partial map w from ports to ports that is symmetric ($w(p) = q$ implies $w(q) = p$), recording which port is joined to which by plain track; a port with no partner is an unwired end. A layout *uses only switches* $0, \dots, N - 1$ when every wired port is smaller than $3N$.

Nothing forbids joining two ports of the same switch (the lobes of Section 4), nor $w(p) = p$: a port joined to itself makes the train re-enter the port it has just left, which is exactly a dead end that reverses the train (in the toy, a buffer with a direction-change stone). The upper bound therefore covers reversing terminators, while every attainment layout below uses ordinary two-ended track only.

Definition 2.2 (Tongues, states, the step)**.** A *tongue vector* is a function u from switches to $\{\mathsf{L}, \mathsf{R}\}$. A *state* is a pair (p, u): the train is about to enter port p with tongues u. One *step* first performs the local passage,

$$\mathrm{arrive}(u, p) = \begin{cases} \big(\text{branch of } k \text{ selected by } u(k),\ u\big) & p = 3k \text{ (facing entry)}, \\ \big(3k,\ u[k \mapsto \text{the branch } p]\big) & p \in \{3k{+}1, 3k{+}2\} \text{ (trailing entry)}, \end{cases}$$

producing an exit port and possibly one changed tongue; the train then travels over the exit port's plain-track edge, so the next state is $(w(\mathrm{exit}), \cdot)$ if the exit port is wired and the run *dies* otherwise. We write $\mathrm{step}^k(c)$ for k iterated steps from state c (undefined once the run dies), and call time k *live* if $\mathrm{step}^k(c)$ is defined.

Definition 2.3 (Restricted vectors and the exact count)**.** For a run from c_0 on a layout using switches $0, \dots, N-1$, the *restricted tongue vector at time* k is the list $\big(u_k(0), \dots, u_k(N-1)\big)$ of the first N tongues at time k. For $N, c \in \mathbb{N}$, say c is the *exact state count* for N (`IsExactStateCount` N c) if

- (*upper bound*) for every wiring on switches $0, \dots, N - 1$, every start, and every duplicate-free list of live sample times whose restricted vectors are pairwise distinct, the list has length at most c; and
- (*attainment*) some wiring, start, and such a list attain length exactly c.

Theorem 1.1 is then literally $\forall N$, `IsExactStateCount` N $(\min(2^N, N+4))$, with no side condition: $N = 0$ is witnessed by the empty layout.

Remark 2.4. Sampling distinct vectors at chosen live times is the right phrasing of "the run exhibits c distinct vectors": it makes the upper bound a statement about every finite observation of the run and the lower bound an explicit finite certificate.

3 The ceiling: $f(N) \le 2^N$

Proposition 3.1 (Finite-state ceiling; `state_law_two_pow`)**.** *For every wiring, every start, and every list of sample times whose restricted tongue vectors are pairwise distinct, the list has length at most 2^N. (No liveness assumption is needed.)*

Proof. Send a boolean list $b_0 b_1 \cdots b_{m-1}$ to its little-endian binary value $\sum_i b_i 2^i$. On lists of a fixed length m this map is injective — the head is the value mod 2, the tail is its half, and induction finishes — and its values are smaller than 2^m. Restricted vectors all have length N, so pairwise-distinct restricted vectors map to pairwise-distinct naturals below 2^N, and there are at most 2^N of those. □

The minimum in the state law changes sides at $N = 3$:

Lemma 3.2 (`add_four_le_two_pow`)**.** $N + 4 \le 2^N$ *for every* $N \ge 3$*; hence* $\min(2^N, N+4) = N+4$ *for* $N \ge 3$*, while* $\min(2^N, N+4) = 2^N$ *for* $N \le 2$*.*

Proof. $3 + 4 = 7 \le 8 = 2^3$, and doubling grows faster than adding one. □

4 Attainment for $N \le 2$

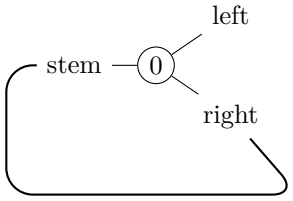
Here $\min(2^N, N+4) = 2^N$: *every* tongue vector must be visited. Each witness below is a finite run; in the formal text these are checked by the proof kernel's `decide`, and the reader can replay them from the tables.

4.1 $N = 0$: the empty layout

With no switches there is one restricted vector, the empty one, and the layout with no track at all exhibits it at time 0. So $f(0) = 1 = 2^0$.

4.2 $N = 1$: the teardrop

Construction 4.1. Wire the stem of switch 0 to its own right branch ($w(0) = 2$), and leave the left branch unwired.



stem–right teardrop; left branch capped

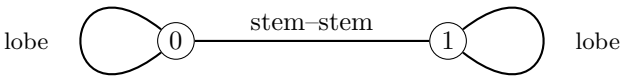
Start the train entering the right branch (port 2) with tongue L. The trailing entry pins the tongue to R and exits the stem; the stem's edge leads back into port 2. The two sampled times already show both vectors:

time	entering port	tongue of switch 0
0	2	L
1	2	R

So $f(1) = 2 = 2^1$. (Thereafter the vector stays R: the run is live forever but mints nothing new — a first taste of the upper bound.)

4.3 $N = 2$: the dogbone Gray square

Construction 4.2. Join the two branches of switch 0 to each other ($w(1) = 2$), the two branches of switch 1 to each other ($w(4) = 5$), and the two stems to each other ($w(0) = 3$).



Each branch-to-branch loop is a *lobe*: a train leaving the stem comes around the lobe and re-enters through the *other* branch, trailing, which flips that switch's tongue — lobes alternate. Start the train entering the stem of switch 0 with both tongues L (write a vector as the pair $u(0)u(1)$). Every second step completes one lobe pass and flips one tongue, walking the full Gray cycle of the 2-cube:

time	entering port	vector
0	0 (stem 0)	LL
2	3 (stem 1)	RL
4	0 (stem 0)	RR
6	3 (stem 1)	LR

(The odd times enter the lobes' far branches: ports $2, 5, 1, 4$.) All four vectors of two switches appear, so $f(2) = 4 = 2^2$. The next flip returns to LL at time 8: the run cycles through the square forever.

5 Attainment for $N \geq 3$: the extremal family

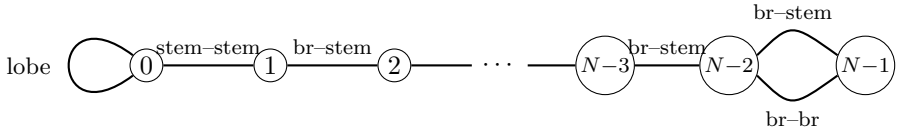
Now $\min(2^N, N + 4) = N + 4$, and one explicit family of layouts attains it for every $N \geq 3$ (`state_law_lower_bound`, file StateLawLowerBound.lean; found empirically by a Python state-count prober, then proved symbolically in N).

Construction 5.1 (The extremal wiring)**.** On switches $0, \ldots, N - 1$:

- switch 0 carries a lobe (its two branches joined, $w(1) = 2$), and its stem is joined to the stem of switch 1 ($w(0) = 3$) — together a teardrop hanging at the bottom;

- switches $1, \ldots, N-3$ form a chain: the right branch of switch k is joined to the stem of switch $k+1$;
- switches $N-2$ and $N-1$ are doubly linked: the left branch of $N-2$ to the stem of $N-1$, and right branch to right branch.

All other ports are unwired.



Theorem 5.2 (Lower bound, $N \geq 3$)**.** *On the wiring of Construction 5.1, the cold start — entering the right branch of switch $N-2$ with all tongues* L *— is live at the $N+4$ times*

$$0,\ 1,\ \ldots,\ N-2,\quad N,\quad 2N-1,\quad 2N,\quad 3N-1,\quad 4N-1,$$

and the restricted tongue vectors at those times are pairwise distinct. Hence $f(N) \geq N+4$.

Proof (rendered). Identify a vector with its set of R switches. The run has four phases, and the sampled vectors are:

time	vector (R-set)
$0, 1, \ldots, N-2$	$\varnothing,\ \{N{-}2\},\ \{N{-}3, N{-}2\},\ \ldots,\ \{1, \ldots, N{-}2\}$
N	$\{0, 1, \ldots, N{-}2\}$
$2N-1$	$\{0, \ldots, N{-}1\}$
$2N$	$\{0, \ldots, N{-}1\} \setminus \{N{-}2\}$
$3N-1$	$\{0, \ldots, N{-}1\} \setminus \{N{-}2, 0\}$
$4N-1$	$\{0, \ldots, N{-}1\} \setminus \{0\}$

Phase 1 (descent, times $0, \ldots, N-2$). The cold train enters the right branch of $N-2$ trailing, pinning $N-2$ to R and leaving by its stem, which is wired to the right branch of $N-3$; the trailing cascade rolls down the chain, pinning one new switch per step — each sampled prefix mints a new vector.

Phase 2 (teardrop, times $N-1, \ldots$). From the stem of switch 1 the train reaches the stem of switch 0 facing. Its tongue says L, so the train exits by the left branch into the lobe and comes back through the right branch, trailing: switch 0 flips to R (the time-N vector). Emerging from stem 0 it re-enters stem 1 facing; each chain switch's tongue now points at the branch the train arrived by in Phase 1, so the train retraces the chain *backwards by pure facing moves, changing nothing* (the retrace mechanism of Section 6.1), reaching the stem of switch $N-2$ at time $2N-3$. Its tongue, set to R in Phase 1, sends the train out the right branch, and the right–right link delivers it into the right branch of switch $N-1$ at time $2N-2$.

Phase 3 (closing the far switch). That trailing entry sets $N-1$ to R: at time $2N-1$ all N switches are R. The train leaves by the stem of $N-1$, whose link leads into the *left* branch of $N-2$, trailing — so $N-2$ flips to L at time $2N$ (the train now heading down the chain again).

Phase 4 (the Gray oscillation). From here the run oscillates on the two coordinates $N-2$ and 0 only: down the chain (trailing entries that change nothing), around the teardrop lobe —

which flips switch 0 — back up by facing retrace, and around the double link — which flips $N-2$: the tongue of $N-2$ decides whether the train arrives at $N-1$ through its stem or its right branch, and either way it returns into the other branch of $N-2$. Concretely, 0 flips to L at time $3N-1$, $N-2$ back to R at $4N-1$, 0 back to R at $5N-2$, and so on with period $3N-1$: the four corners of the square are visited in Gray order, and times $3N-1$ and $4N-1$ sample the two corners not yet seen.

Distinctness is a matter of pointing at one differing coordinate for each pair, and the table makes the witness explicit: within Phase 1, coordinate $N-1-m$ separates the m-th prefix from the later ones; coordinate 0 separates Phase 1 from time N; coordinate $N-1$ separates times $\leq N$ from times $\geq 2N-1$; and the four late vectors differ on $\{N-2, 0\}$, where they realise the four combinations. (In the formal text the trajectory claims are proved by induction on each phase and the liveness and distinctness checks are explicit; the $N=3$ base case, where the chain is empty, is checked by the kernel.) □

6 The sharp upper bound: $f(N) \leq N+4$

This is the mathematical core. The formal statement (`state_law_N_add_four`) is exactly the upper-bound half of Theorem 1.1; its proof occupies most of the development. What follows is a faithful map of that argument: every definition and numbered claim below corresponds to a formal one, and the file named in brackets is where it lives. Proofs here are proof *sketches* — the intended level is "the reader sees why it is true and what must be checked"; the formal text is the complete verification.

Throughout, fix a wiring w on switches $0, \ldots, N-1$.

6.1 Trailing cascades and the retrace lemma

The engine of everything is the asymmetry of the lazy point: trailing passes write, facing passes read.

Lemma 6.1 (Cascades are wiring-determined; GeneralN.lean)**.** *A trailing entry exits by the stem regardless of the tongues. Hence a maximal run of consecutive trailing passes (a* cascade*) — its route, and the pins it writes — depends only on the wiring and the entry port, not on the tongue vector.*

Lemma 6.2 (Retrace; GeneralN.lean)**.** *After a cascade, enter the* stem *of its last switch with any tongue vector agreeing with the cascade's pins. Then the walk performs pure facing moves that traverse the cascade backwards, touches no tongue, and emerges over the cascade's original entry edge.*

Sketch. Each cascade pass pinned its switch's tongue toward the branch the train entered by. Arriving later at that switch's stem, the tongue therefore selects exactly that branch: the train is routed back along the edge it came in by, facing, changing nothing. Induction along the cascade. □

This one-switch fact — a passage leaves the points set so that immediately traversing it backwards is a no-op (`arrive_back`, TrackTrace.lean) — is why grooves, teardrops bounce, and long paths are re-walked for free. We say a path is *grooved* at a vector u when every passage of the path is the one u selects, and call a path *switch-simple* if it visits each switch at most once.

A switch-simple path has at most N passages, and any tongue changes it makes survive to its endpoint.

6.2 First revisit: cycles and manufactured reflectors

Follow the train from its start and record the *physical trace*: the sequence of local passages. Because there are finitely many switches, some switch is eventually revisited; the analysis of the *first* revisit is the fork in the whole proof (TrackTrace.lean, TrackLobe.lean, TripleSelfLinkSimpleCycleClosure.lean).

Lemma 6.3 (First-revisit dichotomy)**.** *Consider the first time the switch-simple initial exploration re-enters a switch it has already visited. Up to the symmetric cases, one of:*

1. *the walk closes a* simple cycle *and settles on it: after a switch-simple transient, the trajectory becomes periodic over a switch-simple cycle whose vector no longer changes; or*
2. *the walk closes a* lobe *(re-enters the old switch by its other branch) or bounces off a* self-link/cap*: the layout has* manufactured a reflector*.*

Definition 6.4 (Manufactured reflectors; TrackTheta.lean)**.** A *manufactured reflector* on entry edge $e \to g$ consists of a switch-simple *runway* from g to a *mouth*, together with either

- (*stay reflector*) a self-linked arm at the mouth — the cap reflects the train back through the branch it came by, re-pinning the same tongue: the local action is the identity; or
- (*flip reflector*) a lobe at the mouth (the *candy*) — the far contact returns the train through the other branch, so each complete traversal flips exactly the mouth switch's tongue (the *action switch*).

In both cases the return journey is the runway retraced backwards (Lemma 6.2): a complete traversal leaves every tongue unchanged except possibly the one action switch, and exits over the entry edge e. The switches of runway and candy are the reflector's *reusable support*; the action switch of a flip reflector is deliberately *not* counted in it.

The state-side consequences are the first two entries of the budget:

Lemma 6.5 (Construction history; FirstReflectorNovelty.lean, SharpStateLawAssembly.lean)**.** *The complete first manufacturing journey — switch-simple exploration, contact, and reverse runway — exhibits at most $N + 2$ distinct restricted vectors: the at most $N + 1$ vectors along the switch-simple exploration (one new tongue write per step at distinct switches, starting from the initial vector), plus at most one for the activated state. The activation itself, however long the runway, contributes at most that single new vector, because the return is a retrace.*

Lemma 6.6 (Two-phase and four-corner laws; ManufacturedPairNovelty.lean, TrackThetaAllTime.lean)**.** *A complete traversal of a manufactured reflector has exactly two tongue phases: the incoming vector, and that vector with the local action applied. For* two *opposite reflectors A, B bouncing a train between them, every vector at every time lies in the four algebraic corners*

$$u, \quad u + A, \quad u + B, \quad u + A + B,$$

whether or not their supports intersect: if the supports avoid each other this is the composed two-phase law, and if one action lands on the other's grooved support the theta *analysis (TrackTheta*.lean) shows the disturbed groove is either repaired in one trailing pass or captured into*

the old mouth, giving pointwise covers by at most three or four vectors. No path length ever enters: the covers are blind to how long the runways are.

The dogbone of Section 4 is the equality case of Lemma 6.6, and the cap-versus-lobe distinction is exactly the sticky/alternating dichotomy visible in the toy.

6.3 The coordinate budget

All counting is funnelled through one frame (TrackFiniteAlternation.lean). Call a live step *productive* if it changes the restricted vector; its *writer* is the switch it enters. Then a run's distinct vectors are at most

$$1 \;+\; \#\{\text{first productive writes}\} \;+\; \#\{\text{repeated-writer novelties}\},$$

where the first term is the initial vector, the second is at most N — one per switch — and a *repeated-writer novelty* is a productive event of an already-productive switch whose resulting vector is globally new. The whole difficulty of the theorem is that repeated writers may in principle keep minting new vectors; the reflector machinery exists to cap them by a constant. The refined bookkeeping never charges the crude N: it charges the switch-simple *exploration* (at most N coordinates, Lemma 6.5), and every later count is charged *into the same N ambient switch coordinates* — reusable support here, productive first writers of a continuation there — plus explicitly named constant corners. Two overlapping histories are never paid twice: when a second manufactured exploration follows a first, the union of the two construction histories, with their shared boundary vector erased, still fits the frame (TwoHistoryUnionCharge.lean, TwoJourneyTailCountSharp.lean).

6.4 The probe tree after one reflector

Assume now the run's *incoming edge is known*: it entered its start over a wired edge (the general start is lifted in Section 6.5). The first-revisit dichotomy (Lemma 6.3) and a second probe of $N+1$ raw steps after the first reflector generate a finite tree of outcomes, each closed by its own count (files PartialSecondRunSharp.lean, PartialSecondRunNAddFour.lean, OneReflectorContinuation.lean, FirstCycleCountSharp.lean):

outcome	count	mechanism
death before any revisit	$\le N+1$	switch-simple prefix
settles on a simple cycle	$\le N+2$	transient + repeat and settled vectors
reflector, then dead probe	$\le N+2$	dead runs cannot damage the support
reflector, then stable cycle	$\le N+3$	joint charge of support and writers
reflector, then changed contact	$\le N+4$	Lemma 6.7
two opposite protected reflectors	$\le N+4$	Lemma 6.8

The two $N+4$ leaves are the genuine frontier (KnownEdgeNAddFourFrontier.lean); everything else in the tree is $N+3$ or less.

Lemma 6.7 (Changed contact; ChangedContactNAddFour.lean, KnownEdgeNAddFourChangedClosed.lean)**.** *Suppose after the first flip reflector the run makes a first support-changing contact. The general changed-support count pays a compressed lead of $N+3$ plus two fresh Gray corners;*

but the reflector's facing action switch is deliberately absent from its reusable support, so unless the strict pre-contact approach productively first-writes that very switch, the action switch is a reserved ambient coordinate, the lead compresses to $N + 2$, and the branch closes at $N + 4$. In the remaining action-written case the local forward tail has only one fresh corner (a runway alternative is killed by the retrace contradiction), which again yields $N + 3 + 1 = N + 4$.

Lemma 6.8 (Protected pair; ProtectedPairNAddFour.lean, PreReturnProtectedRoute.lean, CompleteRepairFour.lean)**.** *Suppose the second probe manufactures a second reflector opposite the first, with each action avoiding the other's support (the* protected pair*). The two construction histories overlap: either the first reflector's omitted facing mouth is the second construction's first productive writer — then its post-write vector duplicates the first reflector's pre-return vector, and erasing the duplicate gives an $N + 2$ construction cover, to which the protected repair tail adds at most two novelties (Lemma 6.6) — or the overlap analysis degrades gracefully through the explicitly enumerated remaining cases (repair leads have two phases, RepairLeadTwoPhase.lean; facing-forward and backward-contact futures cost a constant, FacingForwardNovelty.lean, TrackEarlyRepairConstant.lean, StateLawTwoCandidate.lean). Every case totals at most $N + 4$.*

Packaging the closed tree gives the keystone:

Theorem 6.9 (Known incoming edge; `knownIncomingEdgeNAddFour`, KnownEdgeNAddFourComplete.lean)**.** *If the start is entered over a wired edge, every duplicate-free list of live sample times with pairwise-distinct restricted vectors has length at most $N + 4$.*

6.5 Arbitrary starts and the productive boundary

A general start need not arrive over an edge — the train may simply be placed at a port. Consider its very first passage (StateLawNAddFourTop.lean). If that passage is *quiet* (changes no restricted tongue — a facing entry, or a trailing entry that re-pins), then from time 1 on the run has a known incoming edge, and the time-zero vector equals the time-one vector, so Theorem 6.9 absorbs it. The only danger is a *productive first passage*: the train is dropped into a branch of some switch k_0 and its very first move flips k_0, so the original vector u exists only at time zero, and the shifted run — which starts at u with k_0 flipped, over a known edge — might use its full $N + 4$ budget on top of it.

Proposition 6.10 (Productive boundary; `ProductiveInitialBoundaryNAddFour`, StateLawNAddFourSharp.lean)**.** *The time-zero vector of a productive first passage is never an extra state: the shifted run together with the original vector still fits in $N + 4$.*

Sketch. Suppose adjoining u exceeds the budget. Then the shifted run is *saturated*: it attains all $N + 4$ of its vectors, every one of its live vectors is among the sampled ones, and u itself never occurs on it (BoundaryNAddFourSaturation.lean). Saturation is a very rigid situation, and the geometry of the first journey narrows it to two *savings*, each worth the one missing unit:

- *Absent saving.* The boundary switch k_0 never occurs in the first manufactured exploration. Then k_0 is a genuinely unused coordinate of the first support; if it is also absent from the second construction's productive first writers, adjoining u is paid by this reserved coordinate outright (BoundaryAbsentSecondWriter.lean, BoundaryAbsentProtectedPair.lean).
- *Occurrence saving.* The initially written switch is met again, unchanged, inside the first journey. A canonical such occurrence forces, via the entry-edge symmetry, an empty runway and a literal self-link (BoundaryCanonicalGeometry.lean); a noncanonical one supplies

a *second* repeated vector in the first journey, and erasing both repetitions makes room to insert u into the history at no cost (BoundaryDoubleDuplicate.lean, BoundaryResidualNovelty.lean, BoundaryOccurrenceDamageElimination.lean).

What remains after the savings are spent are the support-damage outcomes, and both expose *the same* sharp changed contact of the first reflector — the damaged stable cycle through its lead, the damaged opposite reflector through its exploration. Under the absent saving, the shifted run begins at the boundary switch's own stem, so switch-simplicity makes k_0 automatically reserved along the strict approach and closes every such contact at $N+3$ (BoundaryChangedContactSaving.lean, BoundaryApproachActionElimination.lean); under the occurrence saving, the occurrence replacement closes the same contact (BoundaryResidualSharpening.lean). Together with the outright eliminations of the short outcomes (a dead second probe and a preserved second cycle are too small to saturate, ProductiveBoundaryNAddFourComplete.lean), every saturated configuration is contradictory, which proves the proposition. □

Theorem 6.11 (Sharp upper bound; `state_law_N_add_four`)**.** *For every wiring on switches $0, \dots, N-1$, every start, and every duplicate-free list of live sample times with pairwise-distinct restricted vectors, the list has length at most $N+4$.*

Proof. Split the samples at time zero. A quiet first passage reduces to Theorem 6.9 directly; a productive first passage is Proposition 6.10. (StateLawNAddFourTop.lean performs this bookkeeping exactly.) □

7 Assembly

Proof of Theorem 1.1. Upper bound: a duplicate-free sample list is bounded by 2^N (Proposition 3.1) and by $N+4$ (Theorem 6.11), hence by their minimum. *Attainment:* for $N=0$ the empty layout, for $N=1$ the teardrop, for $N=2$ the dogbone (Section 4), each attaining $2^N = \min(2^N, N+4)$; for $N \geq 3$ the extremal family (Theorem 5.2) attains $N+4$, which is the minimum by Lemma 3.2. □

8 About the formalisation

The development lives in theory/lean of the repository and builds with Lean 4 (toolchain 4.32.2) and no external library:

```
lake build
```

compiles all 70 files and ends by printing the axiom audit of the single theorem:

`GeneralN.state_law` depends on axioms: `[propext, Classical.choice, Quot.sound]`.

Points worth recording:

- *One theorem.* `state_law` in StateLaw.lean is the only headline statement; every other declaration is reachable support — the tree is mechanically verified to be exactly the theorem's dependency closure.

- *Trusted base.* The three axioms are Lean's standard ones. There is no `sorry` and no `native_decide`: the finite witness runs of Section 4 and the $N = 3$ base of Theorem 5.2 are evaluated by the proof kernel (`decide`). No definition uses choice to produce data (the development contains no `noncomputable` declaration); classical reasoning appears only as excluded middle inside proofs.
- *Symbolic in N.* The upper bound and the $N \geq 4$ lower bound contain no finite-instance argument: all inductions are structural in the trace or in N.
- *Provenance.* The extremal family was found by an empirical state-count prober; the geometry of the physical pieces never enters the proofs, which is itself a (machine-checked) modelling statement: plain track only transports the train between ports.

entry point	contents
StateLaw.lean	the theorem and its statement language
StateLawSmallN.lean	2^N ceiling; empty/teardrop/dogbone
StateLawLowerBound.lean	the extremal family, $N \geq 3$
StateLawNAddFourTop.lean	arbitrary-start bookkeeping
KnownEdgeNAddFourComplete.lean	the known-edge $N + 4$ theorem
TrackTrace.lean, TrackTheta.lean	traces, reflectors, theta engine
StateLawAxiomAudit.lean	the axiom audit